Kody Takada

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EDUCATION

Master of Science in Engineering (M.S.E), Computer Engineering
University of Michigan – College of Engineering, Ann Arbor, MI — August 2024 – May 2025
GPA: 3.75/4.00
Area of Specialization: Signal & Image Processing and Machine Learning

Bachelor of Science in Engineering (B.S.E), Computer Science/Minor in Chemistry
University of Michigan – College of Engineering, Ann Arbor, MI — August 2020 – May 2024
GPA: 3.76/4.00

SKILLS

Programming Languages: C++, C, Python, HTML, CSS, JavaScript, Typescript, MATLAB, R, Julia, SQL
Tools: Git, ReactJS, Flask, Django, Hadoop, Jupyter, AWS, Shell Scripting, Jenkins, AGILE, Scikit-learn, NumPy, Microsoft Suite

WORK EXPERIENCE

BNSF — **Fort Worth, TX**
Software Engineer — *July 2025 - Present*

General Motors — **Austin, TX**
Software Engineering Intern — *May 2024 – August 2024*
- Built an internal Python package leveraging GeoPandas and PySpark for efficient geospatial data visualization and analysis, handling over 6 million static datapoints within Jupyter.
- Designed the frontend application using ReactJS integrating cloud-based datasets with customizable base maps, supporting data scientists with comprehensive analytical tools, and reduced commercial licensing costs by $500,000.
- Engaged in biweekly Agile sprints, utilizing iterative planning to enhance workflow efficiency and ensure deliverable completion within cross-functional teams.

Cruiter.AI — **Sunnyvale, CA**
Data Analytics Intern — *May 2023 – August 2023*
- Created a machine learning regression model focused on calculating expected salary of tech candidates using Python based on personal attributes for our clients to find the optimal tech talent.
- Developed a normalization algorithm to join data for 600,000+ item datasets using RegEx and fuzzy matching.
- Analyzed and built figures to model the datasets using Seaborn to determine trends for future data ingestion.

MedStory LLC — **Ann Arbor, MI**
Software Engineering Lead — *August 2023-Present*
- Led a cross-functional team of developers in building a mobile health tracking app, managing task delegation and sprint planning, ensuring balanced workload distribution.
- Spearheaded internal demos and milestone reviews for stakeholders, translating complex technical progress into clear, actionable updates for non-technical audiences.

PROJECTS

Search Engine | *Python, JavaScript, Hadoop, HTML/CSS*
- Designed a scalable search engine in Python, incorporating RESTful API practices to deliver accurate search results based on user queries through page analysis and tf-idf calculations.
- Calculated an inverted index of pages with Hadoop to enable efficient retrieval for improved search performance.

Movie Rating Classifier | *Python, scikit-learn, NumPy, pandas, gensim*
- Analyzed Amazon's movie review dataset to train Support Vector Machines for rating classification, demonstrating expertise in feature extraction, hyperparameter tuning, class imbalance handling, and bias identification.

MedStory | Typescript, Python, React Native, Django, AWS
- Built a mobile-first health tracking app using React Native (TypeScript), secure navigation flows, and real-time data visualization; integrated the frontend with a Django REST API for seamless symptom and medication logging.
- Architected a scalable backend with Django and a MySQL database hosted on AWS EC2, designing normalized relational models, user authentication, and robust API endpoints; implemented CORS, error handling, and efficient client-server communication to support personalized logging and future expansion to multi-user environments.

ACTIVITIES

Sling Health at the University of Michigan, *Managing Director* — *April 2023-Present*
Engineering Center for Academic Success, *Computer Science Tutor* — *June 2023-June 2024*
American Institute of Chemical Engineers, *Secretary* — *August 2021-June 2023*